

REPORT ON Drop-off & Did Not Attend rates

Dataset used: IAPT – Waiting Times

WEEK 3 ----- ANA 2

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DATE: 19 – 21 MAY, 2026

Introduction:

This analysis focused on patient attendance behavior and treatment continuation patterns using the NHS Talking Therapies dataset. The objective of the analysis was to compute DNA (Did Not Attend) rates and identify treatment drop-off patterns during mental health treatment sessions. The study aimed to understand patient progression, attendance rates, and treatment continuation within NHS mental health services using funnel-based visualization techniques.

Dataset Description:

The dataset used in this analysis was obtained from NHS Talking Therapies Monthly Statistics data. The dataset contained treatment activity indicators related to appointments, attendance behavior, referrals, waiting times, and treatment continuation.

Important variables used during the analysis included:

- MEASURE_NAME
- MEASURE_VALUE_SUPPRESSED
- GROUP_TYPE
- ORG_NAME1

The analysis was performed using Python libraries such as Pandas, NumPy, and Matplotlib in Google Colab.

***	REPORTING_PERIOD_START	REPORTING_PERIOD_END	GROUP_TYPE	ORG_CODE1	ORG_NAME1	ORG_CODE2	ORG_NAME2	MEASURE_ID	MEASURE_NAME	MEASURE_VALUE_SUPPRESSED
77	2026-03-01	2026-03-31	England	all	all SubICBs	all	all Providers	M084	Count_ApptsDNA	59816
79	2026-03-01	2026-03-31	England	all	all SubICBs	all	all Providers	M086	Count_ApptsAttended	545065
108	2026-03-01	2026-03-31	England	all	all SubICBs	all	all Providers	M184	Count_SecondTreatment	54683
294	2026-03-01	2026-03-31	CommissioningRegion	InvalidRegionCode	Unknown	all	all Providers	M084	Count_ApptsDNA	615
296	2026-03-01	2026-03-31	CommissioningRegion	InvalidRegionCode	Unknown	all	all Providers	M086	Count_ApptsAttended	8350

Methodology:

The analysis began by importing the NHS Talking Therapies dataset into Google Colab. Relevant attendance and treatment progression measures were filtered from the dataset.

The selected measures included:

- Count_ApptsAttended
- Count_ApptsDNA
- Count_SecondTreatment

The dataset values were converted into numeric format and missing values were removed to improve analysis accuracy.

A summary table was then created to calculate overall attendance counts, DNA counts, and treatment continuation counts. Visualization techniques such as funnel charts were used to represent treatment progression and patient drop-off behavior.

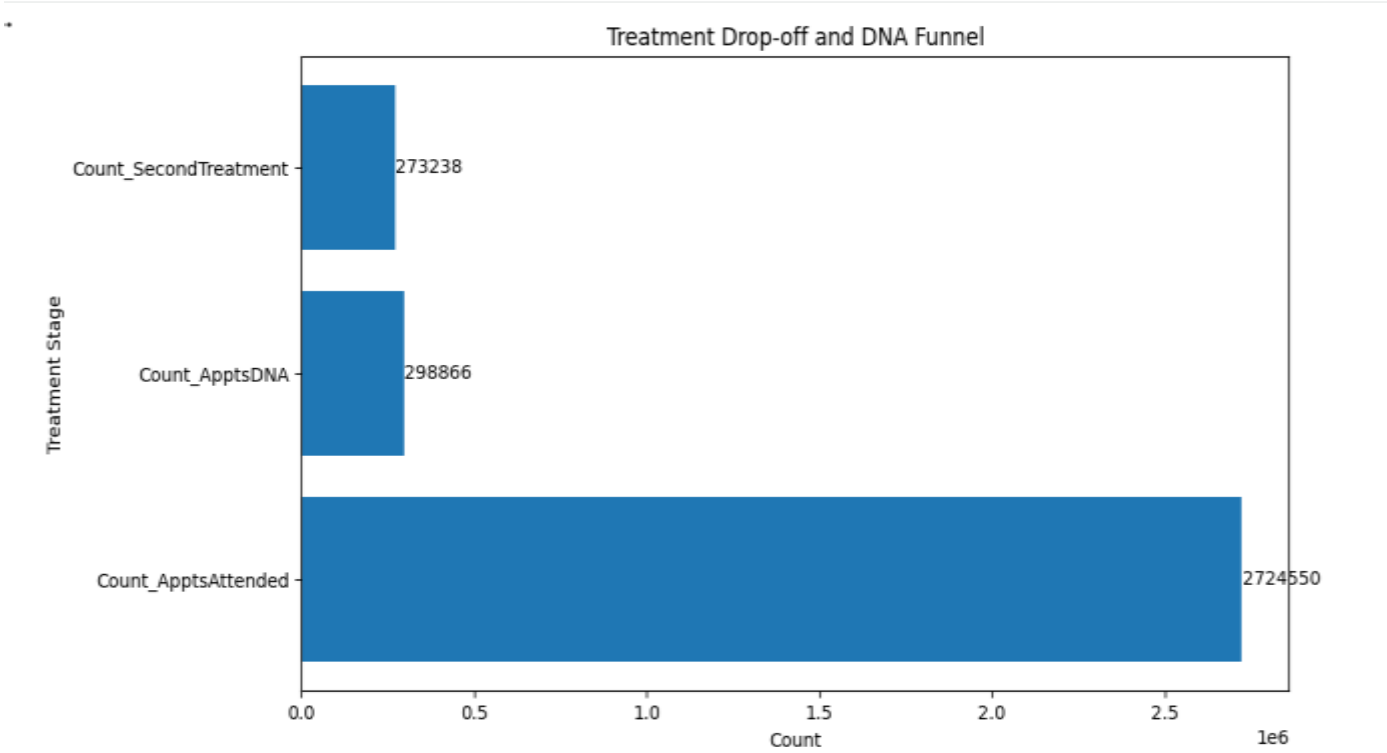
DNA and Treatment Drop-off Summary Table:

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	Measure	Count
0	Count_ApptsAttended	2724550.0
1	Count_ApptsDNA	298866.0
2	Count_SecondTreatment	273238.0

The summary table represents the total number of attended appointments, DNA cases, and second-treatment sessions identified within the NHS Talking Therapies dataset. The analysis highlighted significant variation between attendance and non-attendance patterns during mental health treatment progression.

Treatment Drop-off and DNA Funnel Chart:



The funnel chart represents patient attendance behavior and treatment continuation within NHS Talking Therapies services. The analysis showed that attended appointments were significantly higher than DNA cases and second-treatment sessions. The chart also highlighted patient drop-off patterns during mental health treatment progression. The visualization provided insight into treatment engagement and appointment attendance trends across NHS mental health services.

Key Findings:

1. Attended appointments were significantly higher than DNA cases within the dataset.
2. A noticeable number of patients did not attend scheduled treatment appointments.
3. Treatment continuation into second-treatment sessions was lower than total attended appointments, indicating patient drop-off during treatment progression.
4. The funnel visualization effectively represented attendance patterns and treatment progression stages.
5. The analysis highlighted the importance of patient engagement and attendance consistency within NHS Talking Therapies services.

Conclusion:

The DNA and treatment drop-off analysis successfully identified patient attendance patterns and treatment continuation trends within NHS Talking Therapies services.

The study demonstrated that attendance rates were substantially higher than DNA cases, while treatment continuation decreased during later treatment stages.

The funnel chart visualization provided a clear understanding of treatment progression and patient drop-off behavior throughout mental health treatment services.

Overall, the analysis provided valuable insight into appointment attendance, treatment engagement, and patient progression within NHS mental health services.